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Peer-Video Modeling: Teaching Chained Social Game Behaviors to Children with ASD

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Abstract There is considerable research focusing on remediating social deficits in children with autism. Learning to engage in childhood games such as Duck Duck Goose with peers fosters a multitude of other positive social outcomes. The present study examined peer-video modeling as an intervention to teach two common childhood social games with two participants on the autism spectrum. The games involved chained gross motor behaviors that were performed in an interactive small group setting. Performance was measured by total percentage of correct responding as outlined in a task analysis of the games. Children's performance of chained gross motor behaviors increased across both social games, suggesting peer-video modeling is an effective method in teaching chained social game behaviors.

Keywords Autism · Peer-video modeling · Gross motor · Chained behaviors · social skills

Teaching social skills is a vital component of intervention for children with autism spectrum disorder (ASD), as those skills can often act as a catalyst to other learning opportunities that occur during ongoing social interactions. Social interactions assist in language development, cooperation, and perspective taking, amid other pivotal skills. Children with ASD often do not demonstrate social skills without specific interventions (MacDonald, Sacramone, Mansfield, Wiltz, & Ahearn, 2009; Reeve, Reeve, Buffington Townsend, & Poulson, 2007). Given the significant social deficits in children with autism, considerable research has investigated interventions to teach discrete social skills, such as social initiations and peer-imitation, as well as more

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complex sequenced social skills, such as pretend play scenarios (for a review, see Bellini & Akullian, 2007). Amongst the different social skills interventions, video modeling has rapidly grown in popularity with a number of empirical studies demonstrating its effectiveness (Bellini & Akullian, 2007; MacDonald et al., 2009; Wang & Koyama, 2014).

Often, social behaviors are complex and require sequences of responses as seen with two children playing with a toy house set. One child may move a character to the outside of the door and say, “Knock knock.” The other child may then move her character to the inside of the door and say, “Who’s there?” This scenario may then lead to the first child stating the character’s name, the second child opening the door, and then both children’s characters engaging in a joint activity such as eating a meal. In this case, the sequence of behaviors can be conceptualized as a chain whereby each response (e.g., knocking on door) is also a discriminative stimulus for another response (e.g., asking who is there).

A few studies have examined the role of video modeling in teaching sequences of play behaviors, or chained social skills behaviors. In one investigation, MacDonald et al. (2009) evaluated the effects of video modeling on reciprocal pretend play in two children with autism. Each participant was paired with a typically developing peer and learned to play with toy sets (e.g., a house). The video model portrayed two adults assuming a role and acting out a scripted sequence of pretend play verbal and motor behaviors. Both children with autism exhibited little appropriate reciprocal play during baseline. Following exposure to video models, both children’s scripted verbalization and play action performances increased. Video modeling appears to be an effective intervention for teaching extended sequences of, or chained, cooperative play behaviors in children with autism.

Reviews of video modeling note that the majority of video modeling studies teach discrete social skills (e.g., initiating social responses, social comments) or functional skills that are chained behaviors (e.g., purchasing behaviors, teeth brushing) (Bellini & Akullian, 2007; Wang & Koyama, 2014). While a handful of studies have investigated the acquisition of play sequences and reciprocal social interactions (e.g., D’Ateno, Mangiapanello, & Taylor 2003; MacDonald et al., 2005; MacDonald et al., 2009; Nikopoulos & Keenan, 2004), most of the evidence that video modeling is effective in teaching chained behaviors comes from studies teaching functional skills, such as setting a table or making a sandwich (Bellini & Akullian, 2007). The context in which chained functional skills occur is very different that of social skills. Functional skills typically occur in controlled environments with limited distractions. Social skills, however, occur in more natural contexts in which behavior is directed toward another person, there are plenty of distractions including the social behavior of partners, and these distractions may interfere with acquiring and demonstrating long sequences of behavior. The application of video modeling to chained social behaviors in a social context warrants further investigation.

The purpose of the present study was to extend the video modeling research on the acquisition of chained social behaviors. We taught children with ASD to engage in social games (Duck, Duck, Goose; Hokey Pokey) that are common in preschool and early elementary school settings. The challenge of learning these games is that they require the child to engage in a specific set of behaviors in a specific order in an environment full of distracting stimuli (accompanying music, peers, etc.). The two social games were played in a small group setting and the children both viewed the

video models together and played the game together with a third peer. Engaging in social games affords a multitude of additional social opportunities for additional skills to be learned from observing and engaging with peers.

Method

Participants

Two children (Sally, a 5 year old girl diagnosed with Pervasive Developmental Disorder-Not Otherwise Specified and Andy, a 6 year old boy with Autism) participated in the current study. A third child on the autism spectrum participated in video modeling with Sally and Andy, however, the parent did not provide consent for use of any information about the child's performance to disseminate for research purposes. The participants attended a social skills group that included other children on the autism spectrum. The social skills group was run by two therapists, one of whom was the first author. The participants had been enrolled in the social skills group for approximately 1 year prior to the study. The social skills group met weekly for one hr, during which the children engaged in a number of activities such as board games, card games, video games, arts and crafts, and various playground activities.

Both participants were selected because they met the following inclusion criteria: **a)** diagnosis on the autism spectrum; **b)** use of multiword sentences with extensive grade level vocabulary; **c)** engaged in generalized imitation and followed simple one to two step commands; and **d)** did not independently engage in group activities that involved chained gross motor behaviors, such as the games targeted in this study. Parent report and therapist observations of participants in the social skills group indicated they met these criteria. Parents provided information confirming diagnosis. Diagnoses had been made by professionals not associated with this research using DSM-IV TR criteria (American Psychiatric Association, 2000). When the social skills group played chained social games such as relay races and the parachute, the therapists had to provide constant physical and verbal prompting to redirect both children to participate.

Andy was an affectionate boy who demonstrated an advanced vocabulary, but seldom initiated any interactions with others, including adults. His interactions and conversations were described as rote and lacking natural spontaneity. Andy preferred to play video games on the Wii™ and iPad© rather than interact with others. He participated in several structured group activities that involved turn taking, such as playing board games.

Sally was a high-energy girl who enjoyed reading, using her vast vocabulary to talk about books she read, and completing puzzles. However, appropriate communication was hindered by delayed echolalia, in which she repeated lines from books, movies, or conversations. She manipulated toys appropriately/functionally. She often engaged in solitary play, but had difficulty sharing and engaging with others as demonstrated by crying and screaming when other children played with objects in which she was interested.

Setting, Interventionists, and Materials

This study took place at a center based program that provided 1:1 applied behavior analytic (ABA) therapy, social skills groups, and parent training. Sally and Andy attended a social skills group at the center.

Participants attended their regularly scheduled social skills group in one therapy room (approximately 28 m²). Participants watched video models and played social games without the rest of their social group peers in a nearby therapy room that contained a TV and DVD player.

Two therapists implemented the intervention in this study. The first (first author) was enrolled in the Applied Behavior Analysis Masters program at Queens College, City University of New York. The second was a Masters level educator who recently completed an advanced BCBA certification program. The first author had known the children for 2 years and the other therapist had known the children for 1 year. The therapists were Co-Directors of the center based program and ran the social skills group the children attended.

The interventionists used video recorders to record all sessions. “Let’s Play” Watch Me Learn© videos and YouTube© videos were utilized for the video models. Video models were approximately 40 s in length. Each video clip showed a social game with several typically developing peers, both boys and girls, the same approximate age as the participants. The participants were not familiar with the peers presented in the videos. The first video clip was of five children (3 girls and 2 boys) playing “Duck Duck Goose” and the second was of six children (5 girls and 1 boy) playing “Hokey Pokey.” The first author chose both social games because they were appropriate activities for the age and skill level of the participants. In the videos, the peer models gathered together in a circle and began the behavior chain as outlined in the task analyses (see [Appendix](#)).

Response Definition and Data Collection

The first author created a task analysis for each social game (see [Appendix](#)). The dependent variable was the percentage of independent correct steps performed of the task analysis for each of the two social games, Duck Duck Goose and Hokey Pokey. Correct responses were behaviors emitted within 5 s of the S^D matching the target listed on the task analysis. Behaviors had to occur in the correct sequence in order to be considered correct. No response, a response other than that on the task analysis, or a response occurring after 5 s elapsed was considered incorrect. “Duck Duck Goose” consisted of 12 steps (game played twice) and “Hokey Pokey” consisted of 19 steps (game played once). Mastery criterion was met when a child demonstrated 90 % or greater independent correct responding across two consecutive sessions during post-video viewing observations.

Experimental Design

A multiple baseline design across two social games was used to assess the effects of peer-video modeling intervention and praise on chained gross motor behaviors during social games (“Duck Duck Goose” and “Hokey Pokey”).

Procedure

Baseline, intervention, and generalization occurred during the participant’s regular social skills group at the center. After arriving and engaging in some of the scheduled

social skills group activities for the day, one of the interventionist brought the children to a different room to play the games for this study. During every session (baseline, intervention, and generalization) the participants first completed a no-video probe to assess performance in the absence of video modeling. During intervention sessions, following the no-video probe, participants watched the video model and completed a post-video probe to measure the immediate effects of the video model plus praise. Interventionists always implemented the “Duck Duck Goose” game first, followed by a 3 min break, and then the “Hokey Pokey” game. At the termination of the second game, the children returned to their scheduled activities for the remainder of the social skills group with their typical peers.

Baseline Baseline consisted of a no-video probe during which the interventionist observed the participant’s playing social games in the absence of video modeling, feedback, or reinforcement. A session began with the delivery of the first S^D in the chained sequence and terminated 5 s after delivery of the final S^D (see [Appendix](#)).

Intervention Intervention sessions began with a no-video probe conducted just as in baseline. Then the child viewed the video model and participated in a post-video viewing probe. To view the video model, the children sat in front of a television playing the video. After watching the video one time, the interventionist conducted a post-video viewing probe that was similar to a no-video probe except it occurred after viewing the video model and the interventionist provided behavior specific praise contingent on correct responses. Then, the children took a 3 min break after which they repeated this sequence with the second game.

The interventionist introduced prompting if a child did not emit the specific targeted response for one of the steps outlined in the task analysis (see [Appendix](#)) after 2 consecutive sessions. Using a least to most prompting procedure, prompt levels progressed from gestural to verbal to partial physical, and, finally, full physical. For example, if the child did not engage in the targeted response to put his/her right hand in (during Hokey Pokey), the interventionist provided a gestural prompt by pointing to the participant’s right hand. Prompted responses were scored as a “P” and calculated as an incorrect response; however, the interventionist provided behavior specific praise contingent on correct prompted responses. Prompting only occurred four times throughout the entire intervention.

Interventionists terminated intervention upon mastery level responding. Mastery was achieved when the child performed at least 90 % of the steps of the total task analysis independently and correctly for two consecutive sessions during post-video probes.

Generalization The first author created a task analysis for another childhood social game, “Ring Around the Rosey” and conducted a baseline or no-video probe of participant performance. This was done in order to ascertain whether acquisition of chained behaviors of two social games would occur outside of trained stimuli. Participants did not view video models related to this social game. The interventionist conducted generalization probes in the same manner as baseline once during baseline and twice during intervention.

Social Validity

One parent of each participant rated statements about the appropriateness of learning social game behaviors, social acceptability of intervention, and social significance of outcomes at the beginning and end of the study (see Table 1). We also examined each participant's attendance as a direct measure of social validity.

Interobserver Agreement

All sessions were video recorded. Thirty percent of the total sessions were examined for interobserver agreement (IOA). The first author used a random number generator to choose specific sessions to be coded for IOA. For each session identified for examination for IOA, the interventionist who did not implement intervention independently viewed the video to assess IOA. An agreement was scored if both observer's recorded the same response on each step of the chain for a social game. Agreement was calculated by dividing the number of agreements by the number of agreements plus disagreements and multiplying by 100. Agreement was 98 % (range, 95 % to 100 %).

Intervention Integrity

To evaluate whether the interventionist completed all intervention procedures as outlined, the same observer who examined IOA also examined intervention integrity for the same 30 % of sessions. She used a separate checklist of interventionist behaviors that included setting up the environment, displaying video models, delivering S^Ds, and

Table 1 Parent Social Validity Score Ratings

Statements		
Pre-intervention	Sally	Andy
I think it is important to teach children group social activities.	5	5
I feel that play skills are pre-requisite skills that help children with autism learn additional social skills.	5	5
Based on the peer-video intervention outline provided, I feel it is an appropriate method to use for a child with autism.	5	5
Post-intervention		
I still feel that play skills are important skills for a child with autism to learn.	5	5
I still feel that play skills are pre-requisite skills that help children with autism learn additional social skills.	5	5
I believe using peer-video modeling was appropriate to teach a child with autism how to acquire group social activity skills.	5	5
I believe my child's group social activity skills increased.	5	5
Attendance Rate		
Percentage of sessions attended	92 %	100%

Note: The numbers in the table represent the ratings provided by the participant's parents. Rating scale ranged from 1 (*strongly disagree*) to 5 (*strongly agree*)

providing feedback. Interventionist behaviors were marked as “yes,” “no,” or “n/a.” Integrity was calculated as the number of steps performed correctly divided by the total number of steps of the intervention components, multiplied by 100. Integrity was 100 % for all components.

Results

Figure 1 displays Andy's performance of both social games. During baseline for “Duck Duck Goose,” Andy's performance was low ranging from 17 % to 25 % correct. When intervention began, Andy's performance increased during post-video probes, reaching 100 % independent correct performance and mastery criterion in 7 sessions (one additional session was conducted post-mastery; mastery criterion was at least 90 % of the steps of the total task analysis performed independently and correctly for two consecutive sessions during post-video probes). No-video probes conducted prior to viewing the video model followed the trend of post-video probe performance, but with more variability, reaching 100 % and meeting mastery criterion in 8 sessions. For “Hokey Pokey,” Andy's performance was low at 20 % during baseline. Post-video probe performance increased to 100 % and met mastery criterion in 5 sessions (one additional session was conducted post-mastery). No-video probe performance also increased, reaching 89 % during the last two sessions. It is important to note that 89 % performance reflects 17 of 19 steps performed correctly (only one step could be performed incorrectly to reach the 90 % or better level of performance). During generalization probes for “Ring Around the Rosey” Andy performed 50 % of the social game during baseline. His performance only increased to 60 % during the generalization probe at the end of intervention.

Figure 2 displays Sally's performance of both social games. During baseline for “Duck Duck Goose,” Sally's performance ranged from 0 % to 8 % correct. When intervention was implemented, Sally demonstrated a stable increasing trend during post-video probes reaching 100 % and mastery criterion in 8 sessions. Performance on no-video probes during intervention also increased, but only reached 83 % correct. For “Hokey Pokey,” Sally's performance was low during baseline, ranging from 0 to 5 % correct. Post-video performance increased to 95 %, meeting mastery criterion in six sessions. No-video probes during intervention show a similar increasing trend, reaching 89 % correct responding, as seen for Andy. Performance on generalization probes for “Ring Around the Rosey” increased from 20 % correct responding during baseline to 50 % correct responding during generalization probes at the end of intervention.

Table 1 displays the results of the social validity questionnaires. Parents indicated strong agreement with all statements about the goals, procedures, and outcomes (upper portion of Table 1). High attendance rates (lower portion of Table 1) demonstrate a strong behavioral correlate with social validity ratings.

Discussion

This study replicates and extends the growing literature on the effectiveness of using peer-video modeling to teach social skills to children with autism (e.g., D'Ateno,

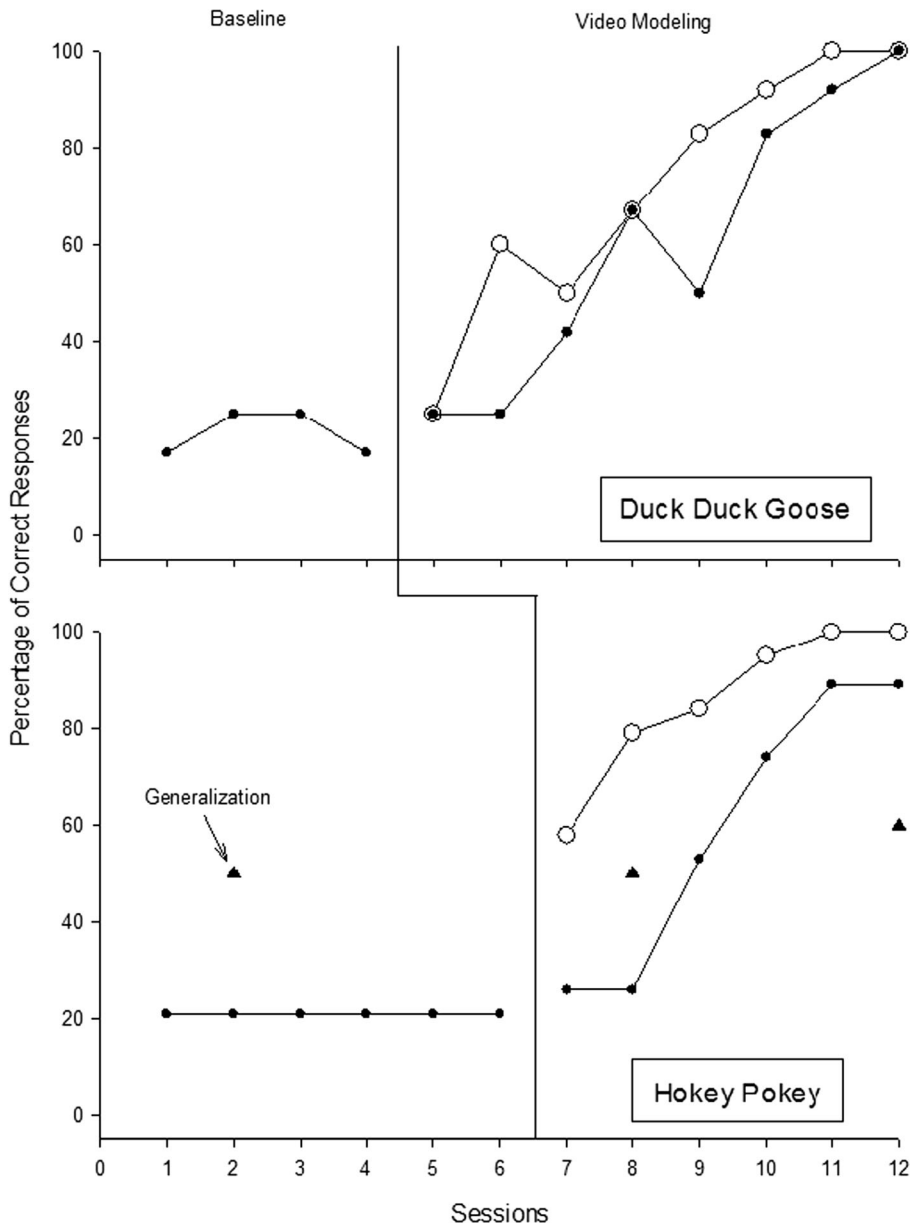


Fig. 1 Percentage of correct responses during no-video probes (filled circles), post-video probes immediately following viewing video models (open circles), and generalization probes (filled triangles) for Andy

Mangiapanello, & Taylor 2003; MacDonald et al., 2009). Both participants' performance of chained social games met mastery criterion after being presented with the peer-video models. No-video probes during intervention revealed an increase in performance in the absence of video models. This is important because post-video observations demonstrate the immediate effects of viewing video models while the no-video probes demonstrate the lasting effect of the video modeling intervention.

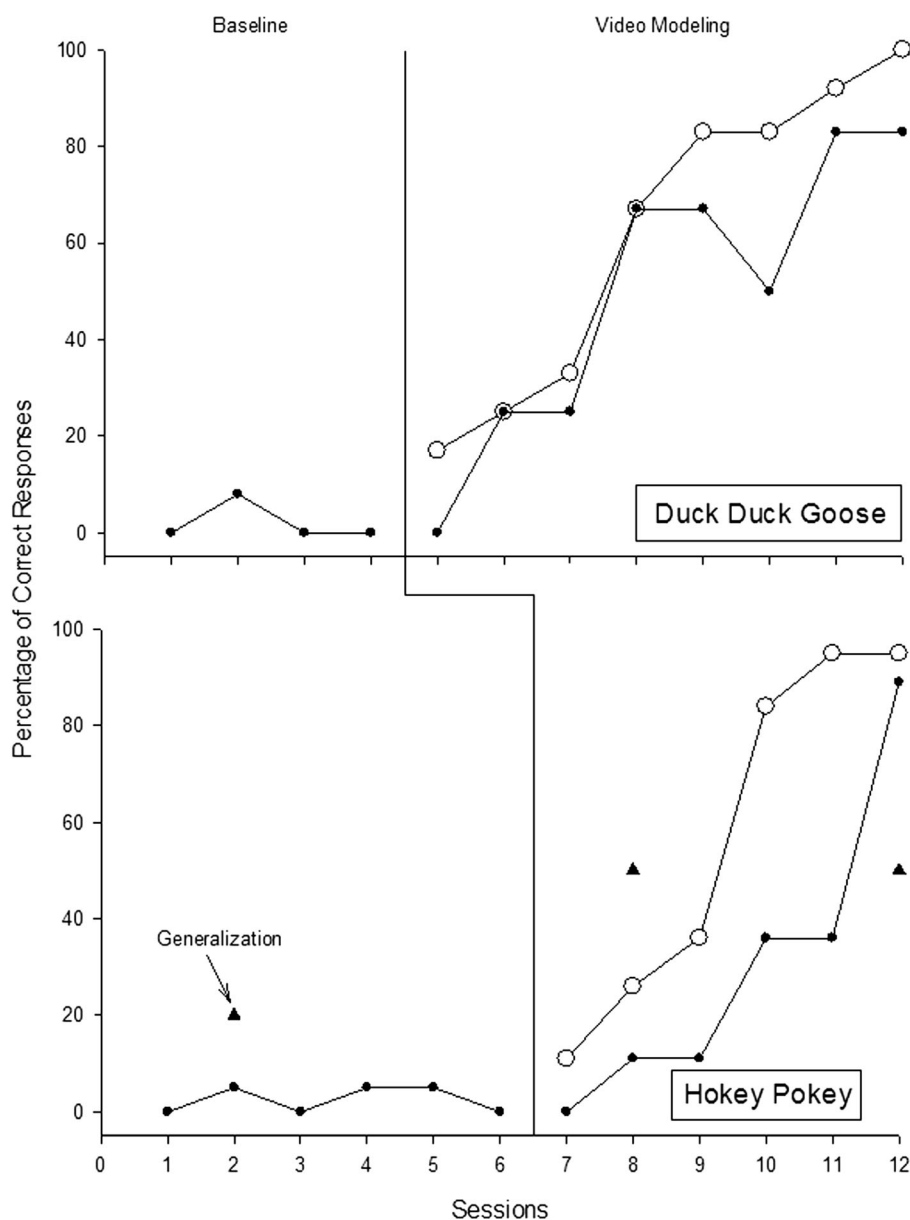


Fig. 2 Percentage of independent correct responses during pre-video probes (filled circles), immediately following viewing video models (open circles), and during generalization probes (filled triangles) for Sally

No-video probe performance scores were slightly lower compared to those collected immediately post-video. Generalization probes revealed a slight increase in performance as well for both participants. Additionally, social validity outcomes were positive. Results were consistent for both parents and attendance rates were also consistent for all participants. The overall findings support peer-video modeling as an effective intervention for teaching children with autism chained social games.

These findings replicate those of MacDonald et al. (2009). The present study and MacDonald et al. both demonstrated the effectiveness of video modeling to teach extended sequences of chained behaviors. Unlike MacDonald et al. who presented a video model with adult models, in this study, videos displayed peers. This does not appear to have negatively impacted performance. The use of peer models may, in fact, may be associated with enhanced performance (Marcus & Wilder, 2009). Choice of peers may affect performance; using familiar and preferred peers might increase attention to the video model and, as a result, positively affect the rate of acquisition and/or occurrence of generalization.

In contrast to other studies (e.g., Nikopoulos & Keenan, 2004), in this study, generalization performance only increased slightly. Other researchers, such as, Reeve et al. (2007) have demonstrated generalization of target behaviors to novel behaviors following video modeling. Their design and procedure provided a multi-exemplar approach, utilizing multiple settings and experimenters, which resulted in all the participants reaching mastery criterion, as well as, generalization to novel settings and people. The present study did not examine generalization to novel settings and people. In this study, “Ring around the Rosey” served as the generalization game. The two games taught through video modeling had little overlap in specific actions to the generalization game. Higher levels of generalization might have been observed with games that included a larger number of overlapping behaviors. As noted previously, choosing peers who are familiar and preferred to show in the videos might also be one way to improve the generalization outcomes observed in this study.

Several limitations exist in the present study. First, the participants selected were chosen based on their current social skill repertoire and deficits. At times, the children were observed attending to stimuli other than the video model. This could have been due to learner preference about viewing videos or something about the videos used in this study. As noted previously, creating videos of familiar and preferred peers might increase attention to the video. In addition, implementing a preference assessment in selecting participants to evaluate whether the participants have a preference for watching television and videos could improve response to intervention. This could provide a good basis for determining preference and potentially reinforcing stimuli, which could ensure greater attending to task. This may also enhance performance results and assist in reaching mastery criterion levels in the absence of video models. It is possible that if trials were extended that all the children could have maintained and carried over post-video performance levels, meeting mastery during the no video probe trials. This could provide a stronger indicator of peer-video modeling effectiveness. Lastly, it is important to note a few potential confounds. The interventionist introduced a least to most prompt hierarchy if a participant did not demonstrate an increase in performance after two consecutive sessions. Prompts were implemented a total of four times during the present study, suggesting a possible confound to video modeling.

This intervention was conducted in a group setting. Each child observed the other children in the group during the activities. Hence, as one participant acquired the targeted behavior, s/he then served as an in-vivo model for other participants above and beyond the video model. For instance, Alex acquired both games quicker than Sally and, thus, Alex may have served as a peer model for Sally. Therefore, in-vivo peer modeling cannot be ruled out as contributing to increased performance. While in-vivo peer modeling is an experimental confound, this also constitutes a practical

strength. This strategy could be used to an interventionist's benefit by layering modeling, such that video models provide a clear and definitive example of target behaviors, but also skilled peers take part in game play to provide an ongoing model of target behaviors. This type of strategy may be well positioned for center and school based programs where instruction occurs with groups of children. Future research should investigate peer video modeling with and without in-vivo peer models to examine whether in-vivo models provide an added benefit to video modeling alone. This line of research could incorporate typically developing peers who have already mastered the specific skills and can serve as peer models.

Appendix

Social Game Task Analyses

Duck Duck Goose:

S ^D	Target Response
1 "Let's play Duck Duck Goose"	Sit in a circle with group
2 Child selected to be "picker" by therapist	Child stands
3 Picker taps one child at a time in clockwise formation around circle while saying "duck, duck"	Children not selected as picker remain seated
4 When picker states "goose"	Child that was taped on head at time of "goose" stands up
5 Picker runs around circle in clockwise formation	Chosen child runs/chases after the picker around the group circle
6 Picker must sit in empty seat before being tagged. If the picker is tagged s/he sits in center of circle. The "goose" is then the new picker and game begins in same fashion.	Picker sits in empty seat or in center of circle. New picker remains standing and begins chain over again.

Hokey Pokey:

S ^D	Target Response
1 "Let's do the Hokey Pokey"	Line up with peers
2 "You put your right hand in"	Place right hand straight out in front of body at shoulder level
3 "You put your right hand out"	Place right hand back at side
4 "You put your right hand in"	Place right hand straight out in front of body at shoulder level
5 "and you shake it all about"	Shake right hand left to right at least 3 times
6 "You do the Hokey Pokey and you turn yourself around"	Turn clockwise 360° twice while raising arms at shoulder level up and down (alternating left and right)
7 "that's what it's all about!"	Back to starting position, clapping hands 3 times
8 "You put your left hand in"	Place left hand straight out in front of body at shoulder level
9 "You put your left hand out"	Place left hand back at side
10 "You put your left hand in"	Place left hand straight out in front of body at shoulder level

	S ^D	Target Response
11	“and you shake it all about”	Shake left hand left to right at least 3 times
12	“You do the Hokey Pokey and you turn yourself around”	Turn clockwise 360° twice while raising arms at shoulder level up and down (alternating left and right)
13	“that’s what it’s all about!”	Back to starting position, clapping hands 3 times
14	“You put your whole self in”	Jump with both feet forward
15	“You put your whole self out”	Jump with both feet backward
16	“You put your whole self in”	Jump with both feet forward
17	“and you shake it all about”	Shake hips left to right at least 3 times
18	“You do the Hokey Pokey and you turn yourself around”	Turn clockwise 360° twice while raising arms at shoulder level up and down (alternating left and right)
19	“that’s what it’s all about!”	Back to starting position, clapping hands 3 times

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